

Testing Services

BOOM SIMULATION report

Oil company: COSMO

Country: U.A.E

Field:

Well:

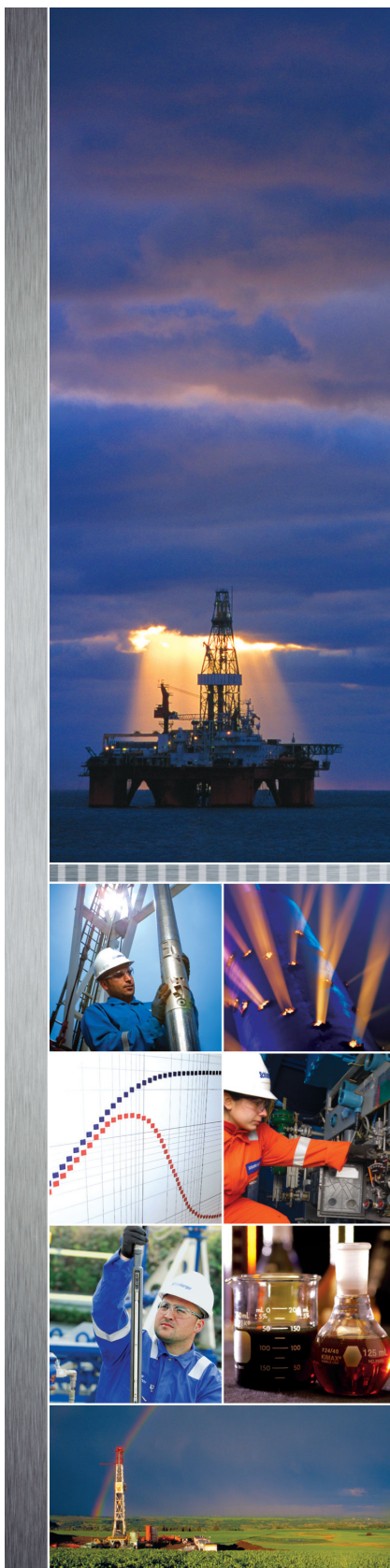
Rig: AL BAZOOM

Date: August 2026

Report number:

Client representative: Avinash Negi

SLB representative: Mahdi Doghri



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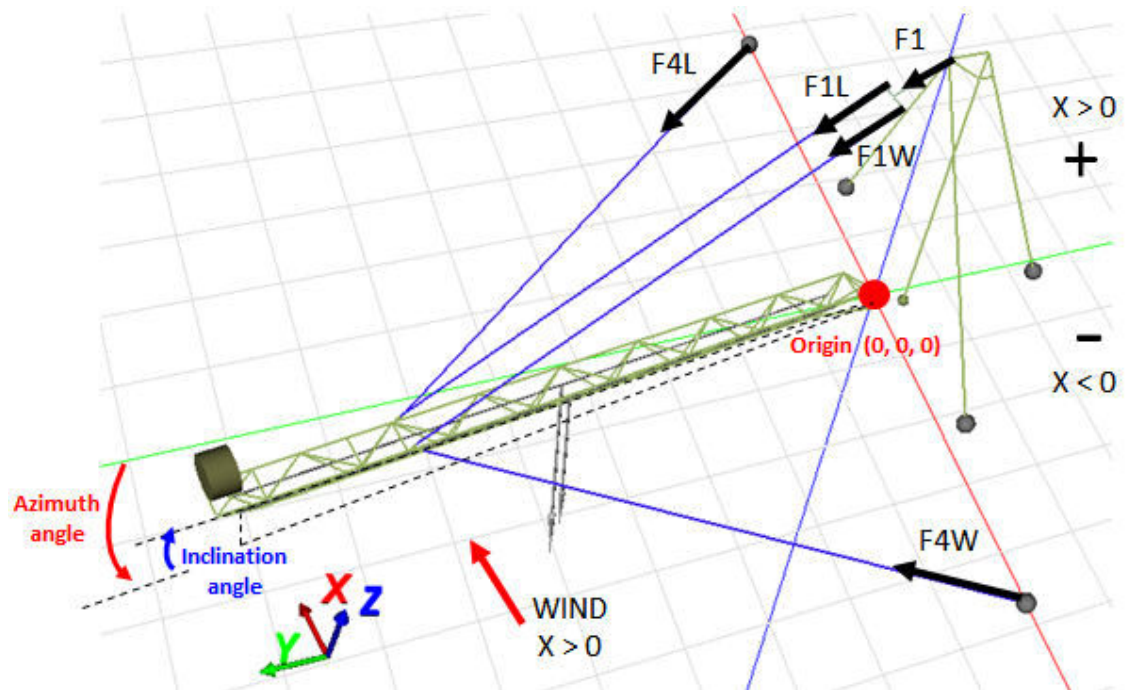
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II. Conventions



III. Inputs

III.1. Equipment list

Boom	Boom asset code	Boom description	Dry weight (kg)	Weight with fluid (kg)	Weight with fluid & ice (kg)	Inclination (deg)	Azimuth (deg)
	UBM-F-60FT		4830.24	5045.01	5045.01	5.00	0.00

Pipes on Boom	Name	Fluid density (kg/m3)	Inner Diameter (cm)	Outer Diameter (cm)
Pipe +	airLine4inches	1.00	10.24	11.43
	gasLine4inches	0.80	10.24	11.43
	waterScreen3inches	1000.00	7.77	8.89
Pipe -	oilLine3inches	850.00	7.77	8.89
	waterLine3inches	1000.00	7.77	8.89
	gasLine4inches	1.00	10.24	11.43

Burner	Burner asset code	Burner description	Weight (kg)	Rotation (deg)
	BRN-HBD		950.00	0.00

Sling attachment points	Name	X (m)	Y (m)	Z (m)
	anchor point hori+	6.43	-2.33	0.10
	anchor point hori-	-6.10	0.80	0.10
	anchor point verti+	0.90	-0.42	12.00
	anchor point verti-	-0.90	-0.42	12.00

Sling	Name	Description	Diameter (cm)	Nominal strength (N)
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	cable hori+	6x37_Class_EIP_IWRC_(6x31,6x36,6x37,6x41)	1.5875	183329.33
	cable hori-	6x37_Class_EIP_IWRC_(6x31,6x36,6x37,6x41)	1.5875	183329.33
	cable verti+	6x37_Class_EIP_IWRC_(6x31,6x36,6x37,6x41)	2.54	460103.21
	cable verti-	6x37_Class_EIP_IWRC_(6x31,6x36,6x37,6x41)	2.54	460103.21

III.2. Additional loads

Name	Load type	X (m)	Y (m)	Z (m)	Resultant (N)	FX (m)	FY (m)	FZ (m)
Water screen + 25%	Node	0.00	13.99	2.33	1404.58	208.10	208.10	- 1373.40
burner + 25%	Node	0.00	17.50	1.53	2668.70	395.39	395.39	- 2609.46
New Node Load_0	Node	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Extra 25 %	Node	0.00	12.16	1.06	6865.58	1017.20	1017.20	- 6713.18

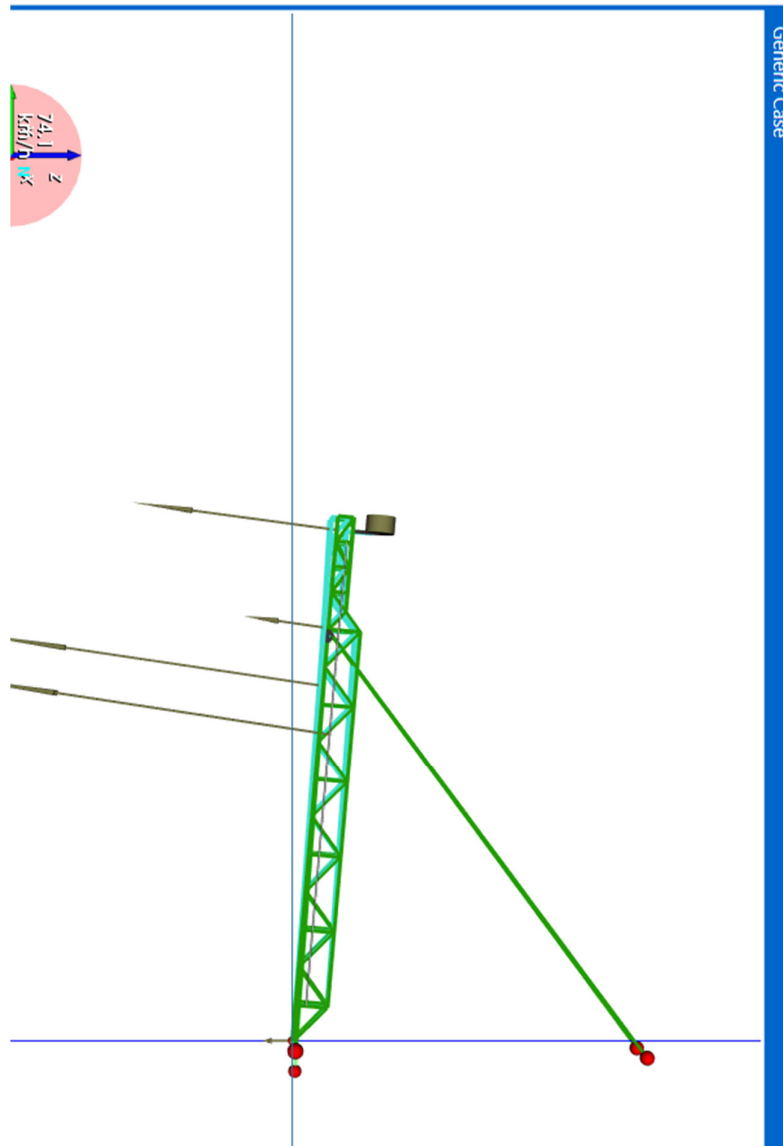
Name	Load type	X (m)	Y (m)	Z (m)	Resultant (N.m)	MX (N.m)	MY (N.m)	MZ (N.m)
Water screen + 25%	Node	0.00	13.99	2.33	0.00	0.00	0.00	0.00
burner + 25%	Node	0.00	17.50	1.53	0.00	0.00	0.00	0.00
New Node Load_0	Node	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Extra 25 %	Node	0.00	12.16	1.06	0.00	0.00	0.00	0.00

III.3. Environment

Rig type	Horizontal acceleration (g)	Vertical acceleration (g)	Ice thickness (cm)	Ice load (kg)	Wind angle (vs X) (deg)	Wind velocity (km/h)	North angle (vs X) (deg)
JACKUP_LOWERED_LEGS	0.24	0.12	0.00	0.00	45.00	74.08	0.00

IV. Results

IV.1. Graphics



IV.2. Alarm color legend

Color	Meaning	Comment
	Not applicable	
	Nil	Null value computed (cable not in tension)
	Low	Computed value less than 80% of the allowable values
	Medium	Computed value between 80% and 99% of the allowable values
	High	Computed value exceed 100% of the allowable values (or missing user input)
	Disable	Alarms disable for the results - Applicable to the anchor points when no information is provided by the client

IV.3. Results on slings

Slings		Tension (N)		Legend
cable hori+	Calculated	0.00		
	User Nominal Strength	183329.33		
cable hori-	Calculated	24308.84		
	User Nominal Strength	183329.33		
cable verti+	Calculated	54266.35		
	User Nominal Strength	460103.21		
cable verti-	Calculated	42351.57		
	User Nominal Strength	460103.21		

IV.4. Results on rig attachment points

The calculated values of this paragraph are only valid for standard rig-up and must be communicated to the rig to confirm that the rig's structure can withstand these forces and torques.

Swivel	Resultant (N)		FX (N)		FY (N)		FZ (N)		MY (N)		Legend
Calculated	89890.58		-2506.10		87360.24		21029.14		-2086.02		
Limit	0.00		0.00		0.00		0.00		0.00		

Attachments		Resultant (N)		FX (N)		FY (N)		FZ (N)		Legend
anchor point hori+	calculated	102.69		-15.21		-15.21		100.41		
	Limit	0.00		0.00		0.00		0.00		
anchor point hori-	calculated	24367.77		-		-		-1740.69		
	Limit	0.00		0.00		0.00		0.00		
anchor point verti+	calculated	54711.58		1183.56		-		32920.46		
	Limit	0.00		0.00		0.00		0.00		

anchor point verti-	calculated	42740.16		- 1473.22		34120.82		25696.71		
	Limit	0.00		0.00		0.00		0.00		

IV.5. Results on structure

(only report medium and high level alarms, if any)

Structure	Number of medium level alarm	Number of high level alarm
Boom	0	11
King post	0	0

Name	Type	Medium Alarms	High Alarms
swivel	REACTION_NODES_BOOM	(none)	Vector Force (high) Vector Moment (high) Force Magnitude (high) Moment Magnitude (high)
anchor point hori+	REACTION_NODES_SLINGS	(none)	Vector Force (high)
anchor point hori-	REACTION_NODES_SLINGS	(none)	Vector Force (high) Force Magnitude (high)
anchor point verti+	REACTION_NODES_SLINGS	(none)	Vector Force (high) Force Magnitude (high)
anchor point verti-	REACTION_NODES_SLINGS	(none)	Vector Force (high) Force Magnitude (high)

Alarms on Boom Structure

IV.6. Load Test Results

V. Appendix

Based on the node displacements, the stresses are computed in all the elements, both at the node locations (ends of the element) and at the middle of the element. Then, those stress values are checked against maximum allowable references. If a stress in an element exceeds the allowable reference, an alarm is raised and the color of the element is modified accordingly. The same approach is performed for the cables.

Concerning the stress reference values, the following norms are considered:

- **API 2A-WSD rev 2007** for cylindrical beam/truss elements.
- **API RP 9B rev 2005** for the cables.
- **AISC steel construction manual 13th edition** for rectangular and I/T beam shapes

The considered stresses are

- Axial tension (beam and truss)
- Axial buckling (beam and truss)
- Bending stress (beam only)
- Shear stress (beam only)
- Torsion stress (beam only)